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United States Patent	12408559
Kind Code	B2
Date of Patent	September 02, 2025
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Semiconductor device and method for fabricating the same

Abstract

A method for fabricating a semiconductor device includes the steps of: forming a first inter-metal dielectric (IMD) layer on a substrate; forming a first metal interconnection and a second metal interconnection in the first IMD layer; forming a channel layer on the first metal interconnection and the second metal interconnection; forming a magnetic tunneling junction (MTJ) stack on the channel layer; and removing the MTJ stack to form a MTJ.

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Appl. No.: 18/587823

Filed: February 26, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240196756 A1	Jun. 13, 2024

Foreign Application Priority Data

CN	202010360779.3	Apr. 30, 2020
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Related U.S. Application Data

continuation parent-doc US 17857185 20220705 US 11950513 child-doc US 18587823
division parent-doc US 16884060 20200527 US 11417838 20220816 child-doc US 17857185

Publication Classification

Int. Cl.: H10N50/01 (20230101); H01F10/32 (20060101); H10N50/80 (20230101)

U.S. Cl.:

CPC H10N50/01 (20230201); H01F10/3254 (20130101); H01F10/329 (20130101); H10N50/80 (20230201);

Field of Classification Search

CPC: H10N (50/01); H10N (50/80); H10N (50/10); H01F (10/3254); H01F (10/329); H01L (21/76838); H01L (21/76877); H01L (23/53209); H10B (61/00); H10B (61/22)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/857,185, filed on Jul. 5, 2022, which is a division of U.S. application Ser. No. 16/884,060, filed on May 27, 2020. The contents of these applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention relates to a semiconductor device and method for fabricating the same, and more

particularly to a magnetoresistive random access memory (MRAM) and method for fabricating the same.

2. Description of the Prior Art

(2) Magnetoresistance (MR) effect has been known as a kind of effect caused by altering the resistance of a material through variation of outside magnetic field. The physical definition of such effect is defined as a variation in resistance obtained by dividing a difference in resistance under no magnetic interference by the original resistance. Currently, MR effect has been successfully utilized in production of hard disks thereby having important commercial values. Moreover, the characterization of utilizing GMR materials to generate different resistance under different magnetized states could also be used to fabricate MRAM devices, which typically has the advantage of keeping stored data even when the device is not connected to an electrical source.

(3) The aforementioned MR effect has also been used in magnetic field sensor areas including but not limited to for example electronic compass components used in global positioning system (GPS) of cellular phones for providing information regarding moving location to users. Currently, various magnetic field sensor technologies such as anisotropic magnetoresistance (AMR) sensors, GMR sensors, magnetic tunneling junction (MTJ) sensors have been widely developed in the market. Nevertheless, most of these products still pose numerous shortcomings such as high chip area, high cost, high power consumption, limited sensibility, and easily affected by temperature variation and how to come up with an improved device to resolve these issues has become an important task in this field.

SUMMARY OF THE INVENTION

(4) According to an embodiment of the present invention, a method for fabricating a semiconductor device includes the steps of: forming a first inter-metal dielectric (IMD) layer on a substrate; forming a first metal interconnection and a second metal interconnection in the first IMD layer; forming a channel layer on the first metal interconnection and the second metal interconnection; forming a magnetic tunneling junction (MTJ) stack on the channel layer; and removing the MTJ stack to form a MTJ.

(5) According to another aspect of the present invention, a semiconductor device includes: a first metal interconnection and a second metal interconnection on a substrate; a first inter-metal dielectric (IMD) layer around the first metal interconnection and the second metal interconnection; a channel layer on the first IMD layer, the first metal interconnection, and the second metal interconnection; and a magnetic tunneling junction (MTJ) on the channel layer. Preferably, a sidewall of the channel layer includes a curve.

(6) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1-6 illustrate a method for fabricating a MRAM device according to an embodiment of the present invention.

(2) FIG. 7 illustrates a 3-dimensional structural view of two MRAM units arranged in an array according an embodiment of the present invention.

(3) FIG. 8 illustrates a cross-section view of FIG. 7 taken along the sectional line BB'.

DETAILED DESCRIPTION

(4) Referring to FIGS. 1-6, FIGS. 1-6 illustrate a method for fabricating a semiconductor device, or more specifically a MRAM device according to an embodiment of the present invention. As shown in FIG. 1, a substrate 12 made of semiconductor material is first provided, in which the

semiconductor material could be selected from the group consisting of silicon (Si), germanium (Ge), Si—Ge compounds, silicon carbide (SiC), and gallium arsenide (GaAs), and a MTJ region **14** and a logic region (not shown) are defined on the substrate **12**.

(5) Active devices such as metal-oxide semiconductor (MOS) transistors, passive devices, conductive layers, and interlayer dielectric (ILD) layer **16** could also be formed on top of the substrate **12**. More specifically, planar MOS transistors or non-planar (such as FinFETs) MOS transistors could be formed on the substrate **12**, in which the MOS transistors could include transistor elements such as gate structures (for example metal gates) and source/drain region, spacer, epitaxial layer, and contact etch stop layer (CESL). The ILD layer **16** could be formed on the substrate **12** to cover the MOS transistors, and a plurality of contact plugs could be formed in the ILD layer **16** to electrically connect to the gate structure and/or source/drain region of MOS transistors. Since the fabrication of planar or non-planar transistors and ILD layer is well known to those skilled in the art, the details of which are not explained herein for the sake of brevity.

(6) Next, metal interconnect structures **18**, **20** are sequentially formed on the ILD layer **16** to electrically connect the aforementioned contact plugs, in which the metal interconnect structure **18** includes an inter-metal dielectric (IMD) layer **22** and metal interconnections **24** embedded in the IMD layer **22**, and the metal interconnect structure **20** includes a stop layer **26**, an IMD layer **28**, and metal interconnections **30**, **32** embedded in the stop layer **26** and the IMD layer **28**.

(7) In this embodiment, each of the metal interconnections **24** from the metal interconnect structure **18** preferably includes a trench conductor and each of the metal interconnections **30**, **32** from the metal interconnect structure **20** includes a via conductor. Preferably, each of the metal interconnections **24**, **30**, **32** from the metal interconnect structures **18**, **20** could be embedded within the IMD layers **22**, **28** and/or stop layer **26** according to a single damascene process or dual damascene process. For instance, each of the metal interconnections **24**, **30**, **32** could further include a barrier layer **34** and a metal layer **36**, in which the barrier layer **34** could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and the metal layer **36** could be selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP). Since single damascene process and dual damascene process are well known to those skilled in the art, the details of which are not explained herein for the sake of brevity. In this embodiment, the metal layers **36** in the metal interconnections **24** are preferably made of copper, the metal layers **36** in the metal interconnections **30**, **32** are preferably made of tungsten or copper, the IMD layers **22**, **28** are preferably made of silicon oxide or ultra low-k (ULK) dielectric layer, and the stop layers **26** is preferably made of nitrogen doped carbide (NDC), silicon nitride, silicon carbon nitride (SiCN), or combination thereof.

(8) Next, a dielectric layer **38** is formed on the IMD layer **28** and a photo-etching process is conducted to remove part of the dielectric layer **38** for forming an opening **40** exposing the metal interconnections **30**, **32** and the IMD layer **28**. In this embodiment, the dielectric layer **38** is preferably formed to accommodate a channel layer formed in the later process so that the thickness of the dielectric layer **38** is preferably maintained between 300 Angstroms to 1000 Angstroms. Preferably, the dielectric layer **38** could include silicon dioxide (SiO₂), silicon nitride (SiN), or silicon carbon nitride (SiCN) and most preferably include SiCN.

(9) Next, as shown in FIG. 2, a channel layer **42** or more specifically a first channel layer **44** and a second channel layer **46** are formed into the opening **40** and on the dielectric layer **38** and a planarizing process such as chemical mechanical polishing (CMP) process is conducted to remove part of the second channel layer **46** and part of the first channel layer **44** so that the top surface of the remaining first channel layer **44** and second channel layer **46** is even with the top surface of the dielectric layer **38** as the first channel layer **44** includes a U-shape cross-section at this stage. In this embodiment, the first channel layer **44** and the second channel layer **46** are preferably made of different materials and/or materials having different etching rates for the ion beam etching (IBE)

process conducted afterwards. For instance, the first channel layer **44** on the lower level is preferably made of material having higher resistance to etching process or with lower etching rate while the second channel layer **46** on the higher level is made of material having lower resistance to etching process or with higher etching rate. In this embodiment, the first channel layer **44** is preferably made of topological insulators including but not limited to for example bismuth selenide ($\text{Bi.sub.xSe.sub.1-x}$) while the second channel layer **46** could include heavy metal such as tantalum (Ta), tungsten (W), platinum (Pt), hafnium (Hf), or combination thereof. Moreover, the first channel layer **44** and the second channel layer **46** preferably have different thicknesses, in which the thickness ratio of the first channel layer **44** to the second channel layer **46** is between 0.5 to 5. For instance, the thinnest portion of the first channel layer **44** is approximately half the thickness of the second channel layer **46** while the thickest portion of the first channel layer **44** is about 5 times the thickness of the second channel layer **46**.

(10) Next, as shown in FIG. 3, a MTJ stack **48** or stack structure is formed on the channel layer **42** and the dielectric layer **38** and a patterned hard mask **50** is formed on the MTJ stack **48**. In this embodiment, the formation of the MTJ stack **48** could be accomplished by sequentially depositing a free layer **52**, a barrier layer **54**, and a pinned layer **56** on the channel layer **42**. Preferably, the free layer **52** could be made of ferromagnetic material including but not limited to for example iron, cobalt, nickel, or alloys thereof such as cobalt-iron-boron (CoFeB), in which the magnetized direction of the free layer **52** could be altered freely depending on the influence of outside magnetic field. The barrier layer **54** could be made of insulating material including but not limited to for example oxides such as aluminum oxide (AlO.sub.x) or magnesium oxide (MgO). The pinned layer **56** could be made of antiferromagnetic (AFM) material including but not limited to for example ferromanganese (FeMn), platinum manganese (PtMn), iridium manganese (IrMn), nickel oxide (NiO), or combination thereof, in which the pinned layer **56** is formed to fix or limit the direction of magnetic moment of adjacent layers. It should be noted that since the present embodiment pertains to fabricating a SOT MRAM device, the free layer **52** is preferably disposed on the bottommost layer to contact the channel layer **42** directly. Preferably, the patterned mask **50** could include conductive material including but not limited to for example metal or metal nitride, in which metal could include titanium (Ti) while metal nitride could include titanium nitride (TiN).

(11) Next, as shown in FIG. 4, one or more etching process is conducted by using the patterned hard mask **50** as mask to remove part of the MTJ stack **48**, part of the second channel layer **46**, part of the first channel layer **44**, and part of the dielectric layer **38** for forming a MTJ **58** on the MRAM region **14**, and a cap layer **60** is formed on the surface of the hard mask **50**, MTJ **58**, second channel layer **46**, first channel layer **44**, and dielectric layer **38**. It should be noted that a reactive ion etching (RIE) process and/or an ion beam etching (IBE) process could be conducted to pattern the MTJ stack **48** and due to the characteristics of the IBE process, sidewalls of the MTJ **58**, sidewalls of the second channel layer **46**, sidewalls of the first channel layer **44**, and top surface of the remaining dielectric layer **38** could all reveal a curve or an arc as the top surface of the part of the dielectric layer **38** could be slightly lower than the top surface of the first channel layer **44**. Preferably, the cap layer **60** could include nitrogen doped carbide (NDC), silicon nitride (SiN), silicon carbon nitride (SiCN), or combination thereof and most preferably SiN.

(12) Next, as shown in FIG. 5, an IMD layer **62** and a stop layer **64** are formed on the cap layer **60**, a planarizing process such as CMP is conducted to remove part of the stop layer **64** and part of the IMD layer **62**, and another IMD layer **66** is formed on the stop layer **64** thereafter. In this embodiment, the IMD layers **62**, **64** preferably include an ultra low-k (ULK) dielectric layer including but not limited to for example porous material or silicon oxycarbide (SiOC) and the stop layer **64** preferably includes nitrogen doped carbide (NDC), silicon nitride (SiN), silicon carbon nitride (SiCN), or combination thereof and most preferably SiN.

(13) Next, as shown in FIG. 6, one or more photo-etching process is conducted to remove part of the IMD layer **66**, part of the stop layer **64**, and part of the IMD layer **62** to form at least a contact

hole (not shown) exposing the hard mask **50**. Next, conductive materials are deposited into the contact hole and planarizing process such as CMP is conducted to form metal interconnection **68** connecting the hard mask **50** underneath, and another stop layer **74** is formed on the surface of the metal interconnections **68** thereafter. Similar to the aforementioned metal interconnections **24**, the metal interconnection **68** could be embedded within the IMD layers **62**, **66** according to a single damascene process or dual damascene process. For instance, the metal interconnection **68** could further include a barrier layer **70** and a metal layer **72**, in which the barrier layer **70** could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (Ta_N) and the metal layer **72** could be selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP). In this embodiment, the metal layer **72** in the metal interconnection **68** preferably includes copper and the stop layer **74** preferably includes nitrogen doped carbide (NDC), silicon nitride (SiN), silicon carbon nitride (SiCN), or combination thereof and most preferably SiCN. This completes the fabrication of a semiconductor device according to an embodiment of the present invention.

(14) Referring to FIGS. **6-8**, in which FIG. **7** illustrates a 3-dimensional structural view of two MRAM units arranged in an array according an embodiment of the present invention, FIG. **6** illustrates a cross-section view of FIG. **7** taken along the sectional line AA', and FIG. **8** illustrates a cross-section view of FIG. **7** taken along the sectional line BB'. As shown in FIGS. **6-8**, the MRAM unit preferably includes metal interconnections **30** and **32** disposed on the substrate **12**, an IMD layer **28** surrounding the metal interconnections **30**, **32**, a channel layer **42** disposed on the IMD layer **28** and the metal interconnections **30**, **32**, and a MTJ **58** disposed on the channel layer **42**. Preferably, the channel layer **42** further includes a first channel layer **44** disposed on and directly contacting two metal interconnections **30**, **32** at the same time and a second channel layer **46** disposed on the first channel layer **44**, a dielectric layer **38** is disposed surrounding the first channel layer **44** and at least a sidewall of the channel layer **42** includes a curve or curves surface.

(15) Viewing from a more detailed perspective, at least a sidewall of the MTJ **58**, at least a sidewall of the second channel layer **46**, at least a sidewall of the first channel layer **44**, and the top surface of the dielectric layer **38** could include one or more than one curve or curved surface, in which the curve of the sidewall of the MTJ **58**, the curve of the sidewall of the second channel layer **46**, the curve of the sidewall of the first channel layer **44**, and the curve of the top surface of the dielectric layer **38** preferably constitute a continuous curve. Moreover, in contrast the bottom surface of the channel layer **42** directly contacting the metal interconnections **30**, **32** in FIG. **6**, the channel layer **42** shown in FIG. **8** only contacts the IMD layer **28** and the IMD layers **62**, **66** and the stop layer **64** are disposed adjacent to two sides of the metal interconnection **68** as the stop layer **64** is embedded in the IMD layers **62**, **66** without contacting the metal interconnection **68** directly.

(16) In contrast to current MRAM devices of utilizing a spin torque transfer (STT) approach for switching magnetic moments, the present invention pertains to fabricate a spin orbit torque (SOT) MRAM device that principally uses SOT effect to switch the magnetic moment within the free layer, or more specifically induces switching of the free layer of the MTJ by injecting an in-plane current in an adjacent SOT layer (or the aforementioned channel layer), typically with the assistance of the state in-plane magnetic field. This enables a three terminal MTJ-based concept that isolates the read/write path, significantly improving the device endurance and read stability.

(17) Under actual fabrication, the present invention preferably employs a damascene process to first form a dielectric layer **38** on the metal interconnections, removes part of the dielectric layer to form an opening exposing the metal interconnections underneath, forms two layers of channel layers made of different materials into the opening, and then forms a MTJ on the channel layers. Preferably, the first channel layer **44** on the bottom is made of topological insulators including but not limited to for example bismuth selenide (Bi₂Se₃) while the second channel layer **46** on the top could include heavy metal such as tantalum (Ta), tungsten (W), platinum (Pt), hafnium

(Hf), or combination thereof.

(18) Since the switching of SOT MRAM is typically achieved by spin current affecting the magnetic moment of the free layer instead of providing current to the device directly, side effect such as reduction of coercivity in the magnetic layers, heating up of the entire MTJ by current, and continuous punch-through of insulating layer could be prevented during write operation of the device. Moreover since the spin current applies equal magnetic field to the entire magnetic layers at the same time so that the chance of switching is only determined by the magnitude of the current pulse applied, it would be desirable to boost up the speed of current plasma applied to the SOT devices than conventional STT devices thereby improving the write speed of the device significantly.

(19) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A method for fabricating a semiconductor device, comprising: forming a first inter-metal dielectric (IMD) layer on a substrate; forming a first metal interconnection and a second metal interconnection in the first IMD layer; forming a channel layer on the first metal interconnection and the second metal interconnection, forming a magnetic tunneling junction (MTJ) stack on the channel layer; and removing part of the MTJ stack and part of the channel layer to form an MTJ, wherein a sidewall of the MTJ comprises a curve.
 2. The method of claim 1, wherein the step of forming the channel layer comprises: forming a dielectric layer on the first IMD layer; removing the dielectric layer to form an opening; forming a first channel layer and a second channel layer in the opening; and planarizing the first channel layer and the second channel layer, wherein the first channel layer comprises a U-shape.
 3. The method of claim 2, further comprising: forming the MTJ stack on the first channel layer and the second channel layer; and removing the MTJ stack, the second channel layer, and the first channel layer.
 4. The method of claim 2, further comprising removing the MTJ stack, the second channel layer, the first channel layer, and the dielectric layer after forming the MTJ stack.
 5. The method of claim 2, wherein the first channel layer and the second channel layer comprise different materials.
 6. The method of claim 2, wherein the first channel layer and the second channel layer comprise different etching rates.
 7. The method of claim 2, further comprising: forming a cap layer on the MTJ, the channel layer, and the dielectric layer; and forming a second IMD layer on the cap layer.
 8. The method of claim 7, wherein the cap layer and the dielectric layer comprise different materials.
 9. The method of claim 1, wherein the MTJ comprises: a free layer on the channel layer; a barrier layer on the free layer; and a pinned layer on the barrier layer.
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